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# AI-based data mining approach to control the environmental impact of conventional energy technologies

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ABSTRACT

Environmental pollution remains one of the foremost existential threats to human well-being, despite the concerted efforts and implementation of various programmes aimed at fostering cleaner air. The contemporary global economic and energy landscape, characterised by multifaceted challenges, has undeniably hindered the efficacy of efforts to kerb air pollutant emissions. Solid fuels persist as primary sources of energy production in numerous countries, serving both the residential and industrial sectors. However, combustion of such fuels, particularly within domestic heating units (DHUs), engenders the release of a diverse array of organic compounds characterised by intricate structures and potent mutagenic and environmentally hazardous properties.

However, the combustion process, if properly regulated, can be carried out in an environmentally sustainable manner. The intricate interplay of myriad factors that influence the composition and quality of chimney flue gases underscores the complexity inherent in controlling the combustion process. Artificial intelligence (AI) has emerged as a versatile tool with applications that span various domains, including environmental monitoring systems. In this study, we posit the utilisation of artificial neural networks (ANNs) as a sophisticated data mining technique to control the emission of flue gases contingent on the specific boiler and fuel utilised. Feed forward predictive models with back propagation were utilized for AI-based data mining aiming at the prediction of the concentration of flue gas components. The highest coefficients of model fit goodness were obtained for $CO_2$, $NO_x$ and $SO_2$ with $R^2$ equal to 0.99, 0.98 and 0.99, respectively. The study demonstrated the feasibility and effectiveness of using AI-based data mining to predict emissions from conventional energy technologies. By leveraging the predictive capabilities of ANNs, it is possible to significantly reduce the environmental impact of solid fuel combustion, contributing to cleaner air and improved public health.

## 1. Introduction

Despite efforts over the past twenty years to reduce the emission of major air pollutants, ambient air pollution remains a critical concern for scientists and policymakers globally, as many regions continue to suffer from substandard air quality. According to the World Health Organisation (WHO), in 2021 in the European Union (EU), 97 % of the urban population was exposed to fine particulate matter concentrations above the level of the health-based guideline. As expected and thanks to numerous programmes aimed at improving air quality, i.e. the zero pollution action plan adopted by the EU in 2021, the WHO has observed a decreasing trend of premature deaths caused by polluted air in 2020 compared to 2005 (*Europe's air quality status 2023*, 2023). The zero pollution action plan sets the target to reduce the number of premature deaths due to exposure to fine particulate matter by 55 % by 2030, compared to 2005. Similarly, the action plan mentioned above sets the 2030 target of a 25 % reduction in the share of ecosystems impacted by air pollution. These assumptions still need the participation of all countries in activities and programmes aiming at improving ambient air quality. The special concern should be still focused on the emission sources particularly polluting the atmosphere, i.e. residential heating.

The ecological requirements for energy-using and energy-related products, including electric and fossil fuelled heating equipment, have been established within the European Ecodesign framework directive (Szramowiat-Sala and Sornek, 2024). However, small combustion devices with the manually controlled (or self-controlled) process of fuel

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**Table 1**
Raw-state results of the elemental analysis of hard coal combusted in the tested boiler.

| Analysed parameters | Symbol | Units | Values |
|---|---|---|---|
| carbon | C | [ %mass] | 67.08 |
| hydrogen | H | [ %mass] | 4.46 |
| nitrogen | N | [ %mass] | 1.36 |
| oxygen | O | [ %mass] | 8.76 |
| sulphur | S | [ %mass] | 0.78 |
| water | W | [ %mass] | 6.50 |
| ash | A | [ %mass] | 11.06 |
| Low calorific value | LCV | [MJ/kg] | 26.51 |

combustion and without any automation are still a popular solution to heat buildings such as holiday cottages, small modular homes year-round, and even one-family homes (Horák et al., 2022; Szramowiat-Sala et al., 2019). If the process of fuel combustion is uncontrolled and if the parameters of this process are not adjusted to the fuel or a boiler, it results in the uncontrolled generation of pollutants and their release into the atmosphere (Rabajczyk et al., 2020). Next to the secondary measures for purification of the flue gas such as oxidation honeycomb catalysts (Ryšavý et al., 2024), utilisation of electrostatic precipitators (Molchanov et al., 2024) and optimization of combustion process conditions could lead to a distinct decrease of emission total suspended particles and polycyclic aromatic hydrocarbons (PAH) (Kirchsteiger et al., 2021). Due to the diversity of the chemical composition of the fuel and thermodynamic parameters, flue gases of different chemical compositions are emitted in the combustion process. In the presence of changing thermodynamic and mechanical conditions occurring in the chimney, flue gas components undergo numerous chemical and physical reactions, e.g. adsorption, dissociation, coagulation, or agglomeration. As a result of these processes, chemical compounds with a high degree of toxicity are formed, which are harmful to humans and the environment (Szramowiat-Sala et al., 2023). Some of the most toxic compounds are PAHs, their nitro and oxygen derivatives, and other compounds with a complex and extensive organic structure (Pacura et al., 2022). These compounds have mutagenic and carcinogenic properties, but are not yet regulated by law.

Artificial intelligence is becoming increasingly applicable in many areas of science, including the environmental and technological disciplines. Deep neural networks (DNN) learn based on the information received, e.g. the operational parameters of the conducted combustion process or the results of concentrations of the flue gas components. A properly trained network can be used to predict the emission levels of pollutants in the flue gas from the combustion process of solid fuels with specific quality parameters and under specific conditions, and finally used for multi-criteria evaluation, which can be considered as analysis of multiple factors influencing flue gas composition, such as different operational parameters of the boiler, which can be further used for the optimization of the boiler's real operational conditions with the aim of decreasing its influence on the environment. The effectiveness of the learning process depends on the amount of training data, the topology of neural networks, algorithms, computational matrices, or the coupling of this tool with other models. The author's literature research (Szramowiat-Sala, 2023) has shown that neural networks have already been used in environmental research, e.g. in the study of industrial emissions, emissions from car engines, or in sewage treatment plants.

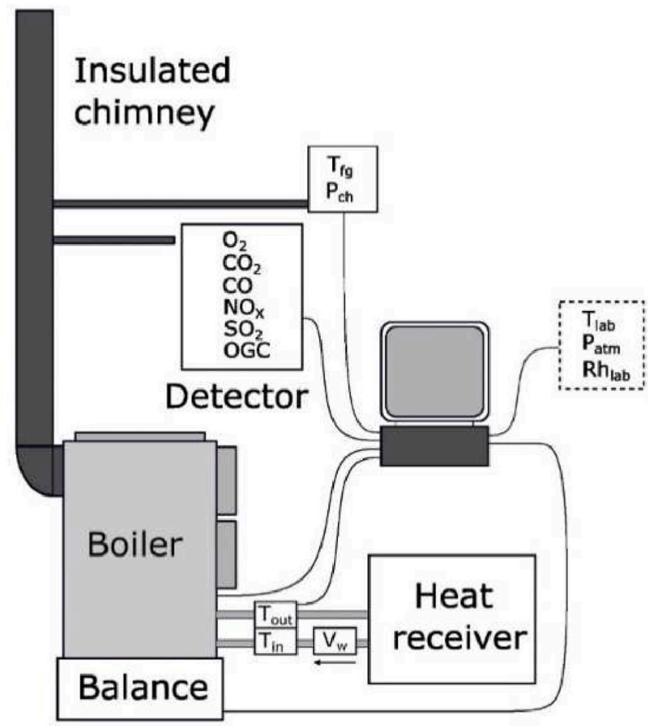


**Fig. 2.** The scheme of the experimental rig (source: own elaboration).

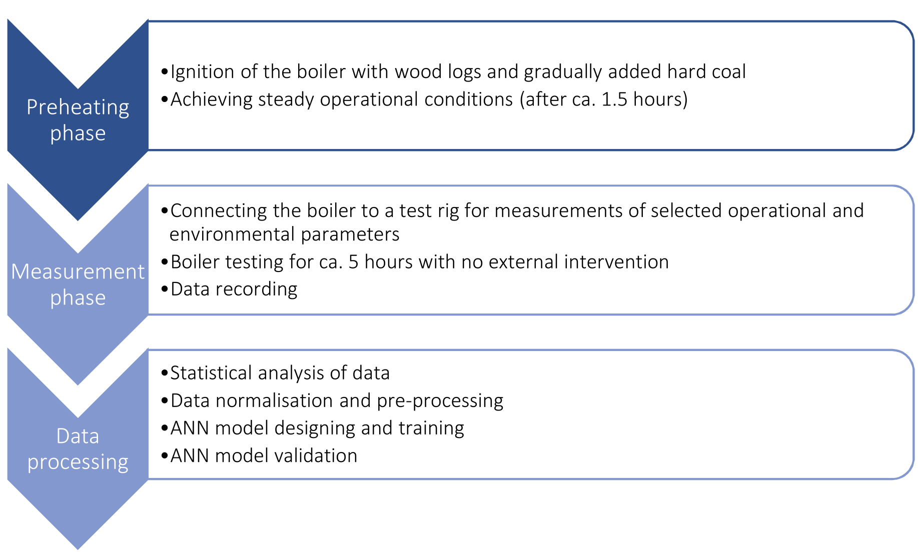


**Fig. 1.** The step-by-step operating procedure of the experimental tests of the gasification boiler used in the study.

**Table 2**
Statistical analysis of data collected during the experimental tests of the gasification boiler.

| | Measured parameter | mean | std | min | 25 % | 50 % | 75 % | max |
|---|---|---|---|---|---|---|---|---|
| **INPUTS** | $T_{IN}$ [°C] | 54.9 | 3.8 | 46.0 | 51.6 | 55.1 | 57.7 | 62.1 |
| | $T_{OUT}$ [°C] | 66.9 | 5.2 | 55.8 | 62.7 | 66.4 | 70.9 | 77.2 |
| | $T_{FG}$ [°C] | 234.5 | 20.6 | 187.5 | 217.2 | 233.4 | 248.2 | 278.5 |
| | $O_2$ [% vol.] | 10.8 | 2.2 | 4.9 | 9.1 | 11.3 | 12.3 | 17.4 |
| | $p_{ch}$ [Pa] | 22.1 | 1.2 | 7.5 | 21.9 | 22.0 | 22.2 | 35.4 |
| | $T_{lab}$ [°C] | 28.5 | 1.0 | 26.2 | 27.7 | 28.5 | 29.2 | 30.3 |
| | $\dot{V}_W$ [$m^3$/h] | 1.5 | 0.00 | 1.5 | 1.5 | 1.5 | 1.5 | 1.5 |
| | $\dot{Q}$ [kW] | 20.3 | 3.1 | 13.6 | 18.0 | 19.2 | 23.1 | 27.3 |
| **OUTPUTS** | CO [mg/$m_N^3$][a] | 1586.0 | 1083.5 | 159.6 | 734.3 | 1297.7 | 2266.8 | 7297.5 |
| | $NO_x$ [mg/$m_N^3$][a] | 428.7 | 253.8 | 105.3 | 236.9 | 347.8 | 505.7 | 1220.7 |
| | $SO_2$ [mg/$m_N^3$][a] | 760.3 | 337.0 | 182.7 | 472.7 | 768.1 | 951.1 | 2078.8 |
| | OGC [mg/$m_N^3$][a] | 124.1 | 166.9 | 2.1 | 4.8 | 30.0 | 226.8 | 961.7 |
| | $CO_2$ [g/$m_N^3$][a] | 175.2 | 36.5 | 72.3 | 148.2 | 169.6 | 206.9 | 259.8 |

'Mean' – arithmetic average, 'std' – standard deviation, 'min' – minimal value of a data set, '25 %' – 25th percentile of a data set, '50 %–50th percentile of a data set, '75 %' – 75th percentile of a data set, 'max' – maximal value of a data set.

[a] The raw data of pollutants concentrations were taken into model considerations. The concentration values of the flue gas components were given under standard conditions: temperature = 0 °C and atmospheric pressure = 101 325 Pa. They were not recalculated to conform Ecodesign Directive requirements.

Regarding residential heating, the application of artificial intelligence for modelling the effectiveness of cogeneration systems (Motahar and Bagheri-Esfeh, 2020), for the prediction of heating energy consumption (Jovanović et al., 2015) of hot water consumption (Maltais and Gosselin, 2021), for minimisation of natural gas consumption of domestic boilers (Tsoumalis et al., 2021), also for optimization of heating and cooling loads in buildings (Li and Yao, 2020; Wang et al., 2022). An interesting solution appears to be an application of ANNs for modelling the relationship between heating energy use and indoor temperatures in residential buildings (Magalhães et al., 2017).

Previous studies have demonstrated the effectiveness of ANN models in predicting and reducing emissions in various boiler types, particularly focusing on industrial large-scale devices. For instance, Wang et al. (2023) utilized convolutional neural network models to predict $NO_x$ emissions in industrial boilers, achieving high accuracy and showcasing the potential for real-time emission control (Wang et al., 2023). Furthermore, Nemitallah et al. (2023) provided a comprehensive review summarizing these studies (Nemitallah et al., 2023). Our research builds upon these foundations by specifically applying ANN models to domestic solid fuel heating boilers under real operating conditions. Unlike previous studies that typically focus on large-scale or laboratory settings, our research aims to offer insights and solutions directly applicable to residential heating systems, which are vital for achieving broader environmental and energy efficiency goals. Additionally, while most reviewed papers concentrate on specific flue gas components (mostly on $NO_x$ reduce), our study offers a comprehensive analysis by considering several flue gas components altogether alongside an examination of the operational parameters of the boiler.

The goal of this article is to develop and evaluate the performance of predictive models based on deep artificial neural networks, assessing the emission characteristics versus operational parameters of a conventional solid fuel boiler. We present the results of implementing ANN in modelling and studying the relationships between emitted flue gas components and the operational parameters of the combustion process in conventional energy technology.

## 2. Experimental study case

### 2.1. Description of conventional energy technology

In experiments a hard coal fuelled gasification boiler was used as an example of conventional energy technology. Fuel gasification boiler is a device that enables the transformation of fuel into gas. It features a fuel storage tank where hard coal undergoes the gasification process. The bottom part of the fuel storage is equipped with fireclay bricks. The gas produced passes through a burner into an insulated combustion chamber, where it is combusted. The resulting flue gases were then directed into a heat exchanger. Both primary and secondary combustion air was preheated. The distribution and amount of primary and secondary air play crucial roles in emissions production. However, due to internal boiler design constraints and narrow air outputs, the primary/secondary air ratio was not measured in the study. The boiler used in the study featured a single central combustion air input, controlled by a flap connected to a thermoelement. This setup modulates air flow based on the output water temperature, adjusting the flap's position throughout the combustion period. The combustion air distribution is managed internally within the boiler, making it impossible to determine the primary/secondary air ratio using mass flow meters. During the experiments, the boiler was set according to the operation manual's recommended values, therefore, the device was used in a typical way for an average user during the tests.

Moreover, the boiler is equipped with a flue gas fan, which is controlled by a control unit (the fan is turned off in case of overheating). The heat output is regulated by adjusting the amount of combustion air. The boiler was certified for the combustion of hard coal and lignite briquettes. Certified nominal heat output of the boiler is 26 kW. In the tests within this document, hard coal was used as a fuel for the gasification boiler. The elemental characteristics of hard coal are presented in Table 1.

### 2.2. The boiler operating procedure and data collection

The experimental tests of the gasification boiler for the purposes of the environmental impact assessment were conducted according to the following step-by-step procedure depicted in Fig. 1.

The boiler was ignited by the wood logs with the gradual addition of hard coal. It took approximately 1.5 h to achieve stable operating conditions, including a stable combustion process, consistent emission values, steady flue gas temperature and heat output. At the end of the ignition and preheating phase, no wood logs were left in the fuel storage and a sufficient basic layer of hot fuel was formed only by hard coal. The measurement phase started immediately after refueling. The measurement phase took about 5 h and no interventions were made in the boiler operation. The experiments were repeated three times.

The boiler was connected to a test rig (Fig. 2). The following operation parameters were registered during each test: inlet water temperature – $T_{IN}$ [$^oC$], outlet water temperature – $T_{OUT}$ [$^oC$], flue gas temperature – TFG [$^oC$], $O_2$ [% vol.], chimney draught – $p_{ch}$ [Pa], atmospheric pressure – $P_A$ [mbar], laboratory temperature – $T_{lab}$ [$^oC$], laboratory relative humidity – $Rh_{lab}$ [ %], water flow in boiler - $\dot{V}_W$ [$m^3$/h], heat output – $\dot{Q}$ [kW]. The concentrations of CO, $NO_x$, $SO_2$, and OGC were given in ppm, resp. in % vol. for $CO_2$ and then transferred to mg/

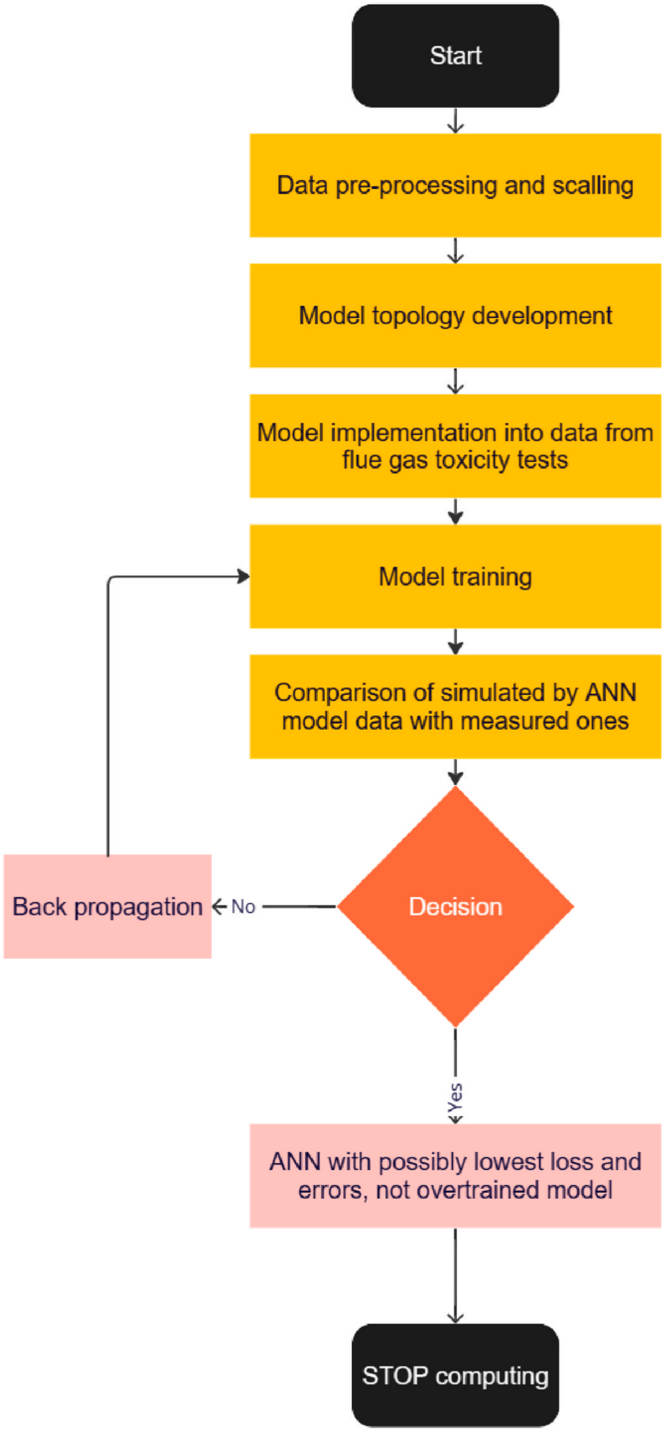


**Fig. 3.** Step-by-step decision making in the data mining approach in this study (source: own elaboration).

$m_N^3$ for CO, $NO_x$, $SO_2$, and OGC, resp. g/$m_N^3$ for $CO_2$.

The water flow was measured using a magneto-induction flowmeter Optiflux 5300 W (Krohne), the water temperatures were measured using paired PT100 sensors (JSP), and the temperature of the flue gas was measured using an OMEGA thermocouple of type K. Fuel consumption ($F_C$) was measured using the Bizerba 2000 VE-M platform balance. Heat output of the boiler was calculated on the base of the water flow. Furthermore, the boiler was connected to a flue gas measurement section that complied with the EN 303–5:2021 test standard (EN303-5:2021+A1:2022, 2021). The flue gas sample was freed of solid particles using a heated ceramic filter and led through a heated sampling line to the flue gas analyser. For the components CO, $SO_2$, $CO_2$ and $NO_x$, the gas sample without solid particles was further dehumidified in the gas cooler and registered by the ABB EL3020 analyser. The concentration of organic gaseous compounds (OGC) in the flue gas was determined using the MultiFID14 analyser (ABB), which works on the principle of a flame ionisation detector. The concentration of $O_2$ in the flue gas was determined using the Magnos206 analyser (ABB), which works according to the paramagnetic principle. The analysers were adjusted with calibration gasses before the tests. The data from the test rig were recorded at 1-s intervals and the average values of 1-min were calculated.

The selection of equipment used during experiments was based on high accuracy and reliability of certain devices in measuring the critical parameters required for our study in order to ensure effective monitoring of combustion process and stable operating conditions. The operation parameters in various configurations served as input for the predictive models, while the emission parameters were treated as separate outputs of these models. Table 2 presents the statistical description of the registered data used as inputs and outputs in the predictive models.

The data presented in Table 2 highlights several key observations. There is significant variability in the emissions of CO, $NO_x$, $SO_2$, and OGC, which could be attributed to fluctuations in the combustion process and fuel quality. The operational parameters such as inlet and outlet water temperatures, flue gas temperature, and chimney draught show relatively stable conditions, ensuring reliable data for the predictive models. The experiments were conducted with the intention of maintaining nominal boiler thermal load; however, there were inherent variations in the actual output due to operational conditions. A gasification boiler with manual refueling tends to exhibit operational periodicity. Based on our experience with hard coal combustion in this boiler, we observed that the heat output gradually increases to its maximum after refueling and then decreases to a minimum. Occasionally, the boiler maintains a relatively stable heat output, but this is not always the case. Table 2 presents statistics from three testing days, which may seem a bit inconsistent. The average heat outputs for the

**Table 3**
Combination of the inputs used and the model for the given predicted output.

| Output | Inputs | Hidden layers architecture | Model fit assessment | | |
|---|---|---|---|---|---|
| | | | $R^2$ | $R^2_{Adjusted}$ | RMSE |
| CO | $T_{IN}$ [°C], $T_{OUT}$ [°C], $T_{FG}$ [°C], $O_2$ [% vol.], $p_{ch}$ [Pa], $T_{lab}$ [°C] | 10/256/10/128[a] | 0.878 | 0.877 | 352.317 mg/$m_N^3$ |
| $NO_x$ | $T_{IN}$ [°C], $T_{OUT}$ [°C], $T_{FG}$ [°C], $O_2$ [% vol.], $p_{ch}$ [Pa], $T_{lab}$ [°C], $\dot{V}_W$ [$m^3$/h] | 10/256/10/128 | 0.980 | 0.980 | 34.022 mg/$m_N^3$ |
| $SO_2$ | $T_{IN}$ [°C], $T_{OUT}$ [°C], $T_{FG}$ [°C], $O_2$ [% vol.], $p_{ch}$ [Pa], $T_{lab}$ [°C], $\dot{V}_W$ [$m^3$/h] | 10/256/10/128 | 0.989 | 0.988 | 34.579 mg/$m_N^3$ |
| OGC | $T_{IN}$ [°C], $T_{OUT}$ [°C], $T_{FG}$ [°C], $O_2$ [% vol.], $p_{ch}$ [Pa], $T_{lab}$ [°C] | 10/4096/128 | 0.903 | 0.903 | 48.204 mg/$m_N^3$ |
| $CO_2$ | $T_{IN}$ [°C], $T_{OUT}$ [°C], $T_{FG}$ [°C], $O_2$ [% vol.], $p_{ch}$ [Pa], $T_{lab}$ [°C], $\dot{V}_W$ [$m^3$/h] | 10/256/10/128 | 0.991 | 0.991 | 3.216 g/$m_N^3$ |

[a] Convention here is assumed that, for example, 10/256/2 means a network with three hidden layers which have 10, 256 and 2 neurones in corresponding layers. The activation function of every layer was ReLU.

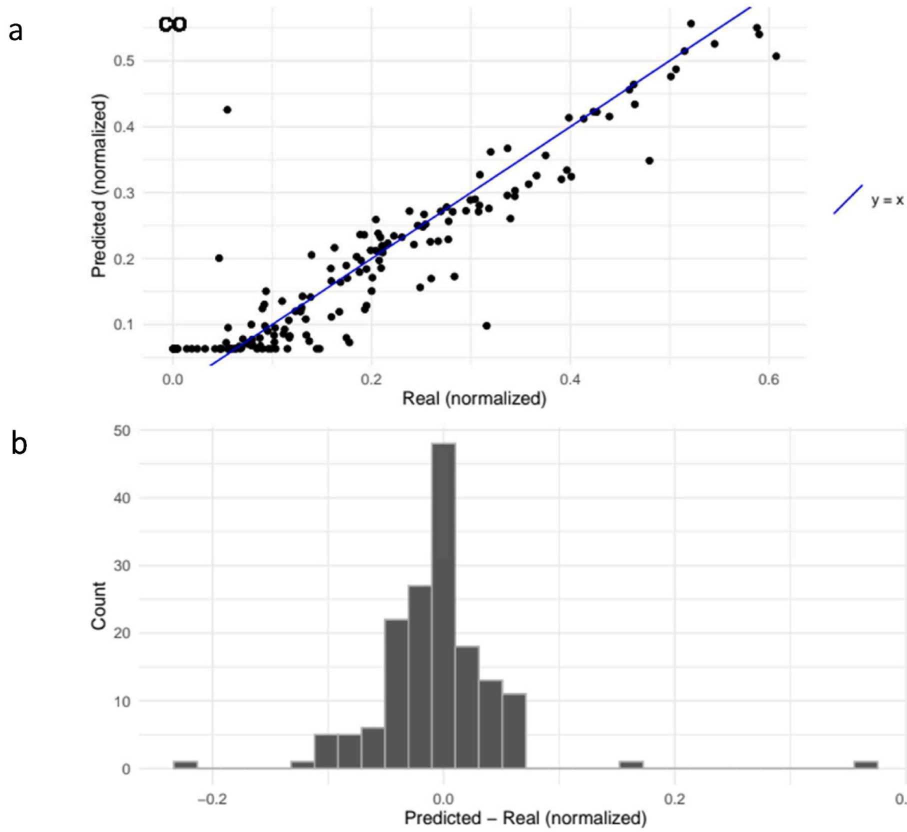


**Fig. 4.** The effectiveness of the ANN model's predictive features to predict CO content in flue gas is demonstrated in the form of (a) CO-predicted vs. CO-measured scatterplot, and (b) a histogram of a difference between predicted and real values of CO concentration (model topology: inputs/10/256/10/128/output).

testing days were 17.9 ± 1.4 kW (n = 323), 23.9 ± 1.8 kW (n = 226), and 20.1 ± 2.3 kW (n = 246). These variations in heat output are due to the boiler's construction—sometimes the fuel ignites better, while at other times, it may get slightly stuck in the fuel storage. It should be remembered that datasets represents results from real-conditions measurements. Despite these differences, the slightly heterogeneous data set is beneficial for training the ANN. This variability helps the ANN models to generalize better and predict emissions more accurately under different operating conditions.

### 2.3. Data mining and model fit assessment

The data mining procedure followed the step-by-step flow chart depicted in Fig. 3. The data treatment process started from a data preprocessing, specifically a stepwise regression, which allowed to identify optimal combinations of independent variable batches as inputs against predicted values of flue gas components as outputs. The results of these computations are discussed in Subsection 3.1. Usually, in most data mining systems, data should undergo the process of transformation. Transformation and normalisation are two frequently used preprocessing techniques. Data transformation revolves around altering raw data input to construct a unified input for a network. In contrast, normalisation focusses on adjusting a singular data input to evenly distribute the data and scale it within a suitable range for network processing. Data transformation encompasses various procedures including normalisation, smoothing, aggregation, and generalisation of data. The analysis of large amounts of complex data can often extend the time required for the analysis to impractical or infeasible durations.

The primary aim of data normalisation lies in ensuring the data quality prior to its utilisation in any learning algorithm. Various forms of data normalisation exist, each serving different purposes. Its utility extends to aligning the data within a consistent range of values across all input features, thus mitigating biases within the neural network from one feature to another. Additionally, data normalisation has the potential to accelerate the duration of training by initiating the training process for each feature within a uniform scale. This proves particularly advantageous in modelling applications, where inputs typically exhibit substantial disparities in scales. Diverse techniques employ different criteria such as min-max, Z-score, decimal scaling, median normalisation, among others (Nayak et al., 2014). For this research, the min-max normalisation technique has been applied. In this technique, the data inputs are mapped into a predefined range [0, 1] or [−1, 1] according to a linear equation (1) (Asami et al., 2021):

$$v' = \frac{v - min_A}{max_A - min_A}(new\_max_A - new\_min_A) + new\_min_A \quad (1)$$

Here $max_A$ is the maximum value of the attribute, $min_A$ is the minimum value of the attribute for ($new\_max_A$-$new\_min_A$) = 0.

The models of artificial neural networks used in this research were sequential models with a few fully connected hidden layers. The activation function of every layer was a Rectified Linear Unit (ReLU). The ReLu function accelerates the training speed of deep neural networks compared to traditional activation functions (like tangh or sigmoid) since the derivative of ReLu is 1 for positive input. ANN models designed in the experiment used regularisation L2 to reduce model overtraining. The models were regression models, i.e. they predicted values based on input values. Each model used the Adam optimiser, which is a popular choice for optimizing weights in neural networks, and was trained for 500 epochs. For each combination, the dataset was split into two parts: 80% of the rows were used to create the training set, and the remaining 20% formed the test set. The training set was used to train the model, while the test set was exclusively used for validation to ensure the model's accuracy and generalization. The loss function was the mean square error (MSE), mathematically expressed in Equation (2):

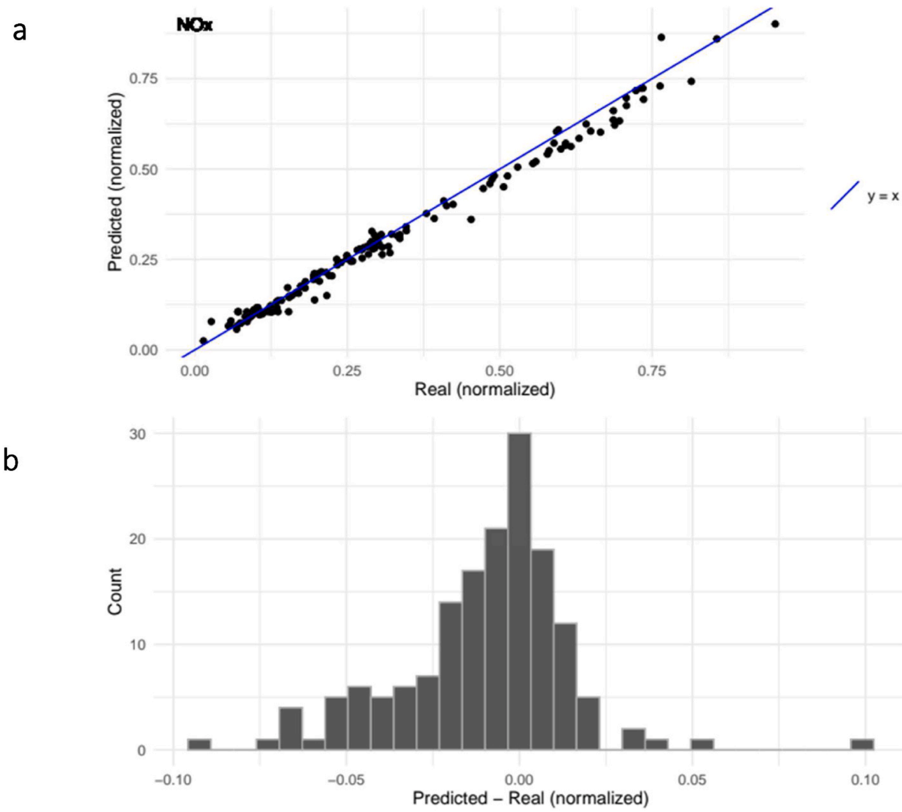


**Fig. 5.** The effectiveness of the predictive features of the ANN model in predicting $NO_x$ content in flue gas is demonstrated in the form of (a) NOx predicted versus NOx measured scatterplot, and (b) a histogram of the difference between predicted and real values of $NO_x$ concentration (model topology: inputs/10/256/10/128/output).

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2}$$

Where: $n$ represents the number of samples in the dataset or observation; $y_i$ denotes the actual target value; $\hat{y}_i$ signifies the predicted value by the algorithm. MSE serves as a loss function that assesses the disparity between the predictions from a predictive algorithm and the actual outputs. Calculates the average of the squared differences between the predictions and the target values. By squaring these differences, more substantial deviations from the target value receive higher penalties. Normalising the total errors against the dataset or observation's sample count is achieved through calculating the mean of the errors. For the purposes of the model fit assessment, three statistical parameters were calculated: coefficient of determination (R2), adjusted coefficient of determination (R2 adjusted), and root mean square error (RMSE) according to Equations (3)–(5):

$$R^2 = \left[\frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2}\sqrt{\sum (y_i - \bar{y})^2}}\right]^2 \tag{3}$$

$$R^2_{Adjusted} = 1 - \frac{(1 - R^2)(n-1)}{n - p - 1} \tag{4}$$

$$RMSE = \sqrt{\frac{\sum (x_i - y_i)^2}{n}} \tag{5}$$

where $x_i$ and $y_i$ – the $i$-th observed and estimated values, respectively; $\bar{x}$ and $\bar{y}$ - the average of $x_i$ and $y_i$; $n$ – the total number of data; $p$ – the number of parameters.

### 2.4. Statistical data analysis

Calculations were done in Python and R with Tensorflow library https://github.com/rocpe/nn-piec. Furthermore, Statistica 13.3 was used. Statistical calculations were performed with a 5-percent level of significance.

## 3. Results and discussion

### 3.1. Models and data preprocessing

To identify the most optimal dependence between boiler operational parameters and concentrations of emitted pollutants and to eliminate those independent parameters with low impact on the modelling process, stepwise regression has been performed. CO, $NO_x$, $SO_2$, OGC, and $CO_2$ were predicted using operational parameters like inlet water temperature, outlet water temperature, flue gas temperature, $O_2$, chimney draught, laboratory temperature, water flow in boiler, and heat output as model inputs in different configurations. The parameters reflecting the fit goodness of the model were calculated for each of 40 input-output combination. The results of stepwise regression have been presented in Table S1. The highest values of $R^2$ (or $R^2_{Adjusted}$) and the lowest values of RSME reflect the best goodness of the model fit. The batch of inputs, which gave the best goodness of model fit, consisted of 6 independent variables (inlet water temperature, outlet water temperature, flue gas temperature, $O_2$, chimney draught, laboratory temperature) for CO and OGC (with $R^2$ equal to 0.854 and 0.930, respectively). For $NO_x$, $SO_2$, and $CO_2$, the batch of inputs that gave the best model fit consisted of 7 independent variables (inlet water temperature, outlet water temperature, flue gas temperature, $O_2$, chimney draught, laboratory temperature, and water flow in boiler) with $R^2$ equal to 0.981, 0.972 and 0.991, respectively.

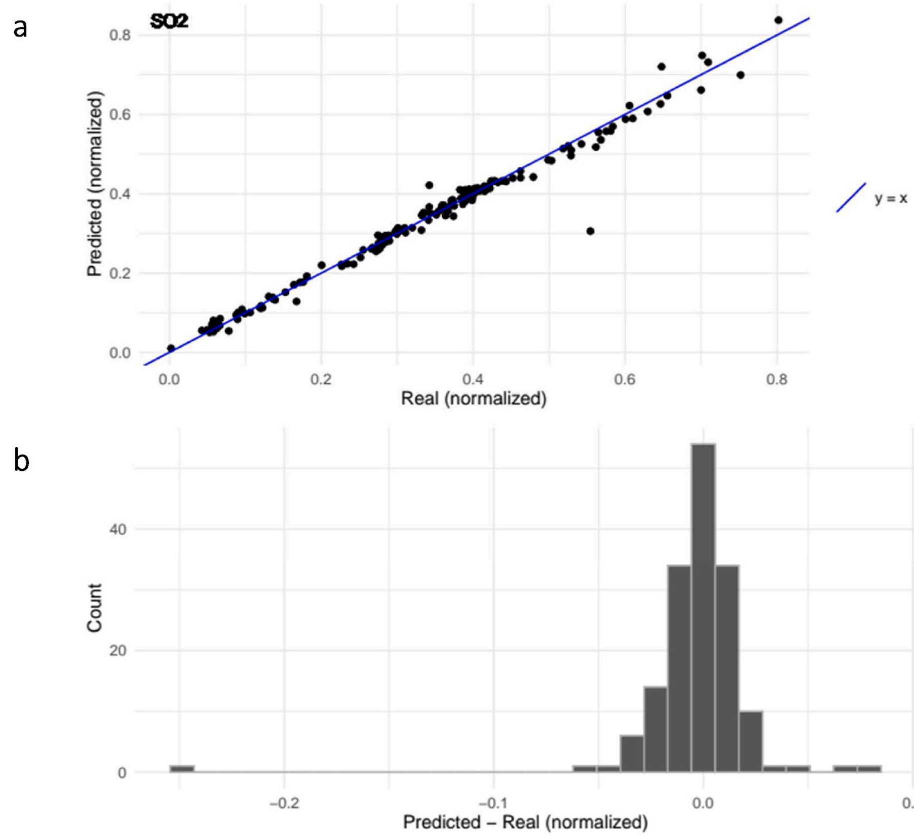


**Fig. 6.** The effectiveness of the predictive features of the ANN model in predicting the $SO_2$ content in the flue gas demonstrated in the form of (a) the SO2 predicted vs. SO2 measured scatterplot and (b) a histogram of the difference between predicted and real $SO_2$ concentration values (model topology: inputs/10/256/10/128/output).

As for the next step, three topologies of the ANN model with an overall structure: inputs/hidden layers/output were tested. The number of neurones in an input layer was either 6 for CO and OGC or 7 for $NO_x$, SO2, and $CO_2$. An output was predicted in each case. The effectiveness in predicting emitted pollutants was checked using three different topologies of hidden layers in artificial neural networks: I. 10/256/10/128, II. 10/2/10/128, III. 10/4096/128. The results of the model fit assessment are summarised in Table S2. For CO, $NO_x$, $SO_2$ and $CO_2$ the architecture of hidden layers assigned as model I and for OGC the architecture no. III demonstrated the highest values of $R^2$ and $R^2_{Adjusted}$ and the lowest values of RSME. The combinations of optimal input variables and hidden layer topology for predicted flue gas components were then extracted and presented in Table 3.

### 3.2. The evaluation of predictive models for boiler emission performance

Figs. 4–8 present: (a) scatterplots of predicted values of flue gas components against measured values, and (b) histograms of differences between predicted and real (normalised) values of CO (Fig. 4), $NO_x$ (Fig. 5), $SO_2$ (Fig. 6), OGC (Fig. 7) and $CO_2$ (Fig. 8).

As observed in the scatter plots, the predictive models employing specific topologies demonstrated a substantial correlation between the predicted and measured concentrations of the flue gas components. This strong correlation is particularly evident for pollutants such as $NO_x$, $SO_2$, and $CO_2$, where the data points closely align with the trend line $y = x$. This alignment suggests that the models were highly accurate in predicting these pollutants' concentrations. The goodness-of-fit indicators for these three pollutants were exceptionally high, with values exceeding 0.98, underscoring the models' robustness and reliability in these cases.

For CO and OGC, although the data points exhibited a greater scatter compared to $NO_x$, $SO_2$ and $CO_2$, the evaluation indicators remained significantly high, approaching 0.9. This indicates that although the models were slightly less accurate for CO and OGC, their predictive performance was still commendably strong. The higher scatter might be attributed to the intrinsic variability in the measurement or prediction processes for these components, or possibly due to the complexity and nonlinear behaviour of their interactions in the flue gas.

To further investigate the performance of the predictive models, "error histograms" were generated (Figs. 4b – 8b). These histograms illustrate the frequency distribution of errors, defined as the differences between the predicted values by the ANN models and the actual measured values of the flue gas components. For CO, $NO_x$, $SO_2$ and OGC, the histograms showed a shape similar to a Gaussian distribution. This bell-shaped distribution is centred around 0.0 (or close to 0.0 for OGC), indicating that most predictions were close to the measured values, with errors distributed symmetrically around zero. This Gaussian-like distribution of errors reinforces the conclusion that predictive models are effective and reliable for these pollutants.

However, the predictive performance of $CO_2$ showed a notable deviation (Fig. 8b). The histogram for $CO_2$ errors did not centre around zero but instead showed a maximum frequency of occurrence near 0.025. This deviation suggests that the predictions for $CO_2$ were consistently slightly higher or lower than the measured values. This pattern could imply a systematic bias in the $CO_2$ model, which could arise from several factors including the nature of the training data, the specific architecture of the ANN, or the inherent complexities in accurately predicting $CO_2$ concentrations.

The study lays the groundwork for further research on the application of AI in environmental monitoring and control systems. Future work could explore more complex models or integrate additional data sources to improve predictive accuracy and operational efficiency.

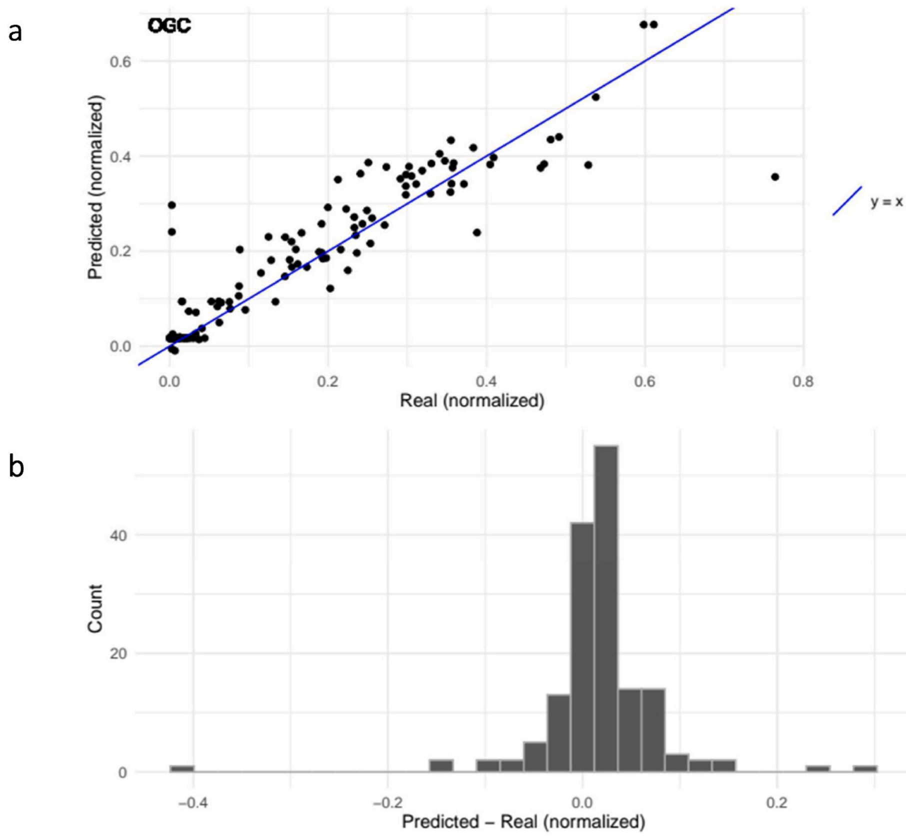


**Fig. 7.** The effectiveness of the predictive features of the ANN model to predict OGC content in flue gas demonstrated in the form of (a) OGC-predicted versus OGC-measured scatterplot, and (b) a histogram of a difference between predicted and real values of OGC concentration (model topology: inputs/10/4096/128/output).

Ensuring the robustness of these models in diverse operating conditions and with different types of fuel is essential for their practical deployment. Implementing AI-based emission control systems could lead to substantial improvements in air quality, particularly in residential areas where small, uncontrolled boilers are still commonly used. Policymakers and manufacturers could leverage these findings to develop smarter heating systems that automatically adjust to minimize emissions. This would contribute to cleaner air and better public health outcomes.

## 4. Conclusions

In the study, the use of artificial neural networks was explored as a sophisticated data mining technique to predict and control emissions based on the specific boiler or boiler and the type of fuel used. ANNs were trained on data from experimental tests of the gasification boiler to assess environmental impact, enabling them to predict pollutant levels under varying conditions. This approach leveraged the versatility of AI in environmental monitoring systems to improve the efficiency and effectiveness of emission control.

The data mining approach presented in this paper provided a comprehensive evaluation of the performance of the ANN models in predicting flue gas component concentrations. The models exhibited high accuracy and reliability for $NO_x$, $SO_2$, and $CO_2$, as indicated by the strong correlations and low error frequencies. For CO and OGC, the models were still highly effective, though with slightly more variability.

The deviation observed in the $CO_2$ predictions highlights an area for potential model improvement, suggesting that further refinement of the model or additional data could enhance predictive accuracy. The relationship found in AI-based data mining between the measured $O_2$ [% vol.] during experiments and the predicted concentrations of flue gas components appears to be the most significant outcome of this research, offering opportunities for optimizing the combustion process to minimize pollutant emissions. These findings demonstrate the substantial potential of ANN models in environmental monitoring and pollutant prediction, providing valuable insights to optimize residential combustion processes and ensure regulatory compliance. This optimization can maintain the highest possible energy efficiency of heat generation systems, benefiting boiler users. Although models show great promise, real-world implementation would require addressing challenges such as data availability, model generalisation, and integration with existing systems. It is crucial to ensure that these models can operate effectively under various conditions and fuel types for them to be viable in practical applications.

## Data availability

The data are not available in open repositories but can be provided upon request from interested parties.

## CRediT authorship contribution statement

**Katarzyna Szramowiat-Sala:** Writing – review & editing, Writing – original draft, Funding acquisition, Data curation, Conceptualization. **Roch Penkala:** Writing – original draft, Data curation. **Jiří Horák:** Methodology, Investigation, Data curation. **Kamil Krpec:** Methodology, Investigation, Data curation. **František Hopan:** Methodology, Investigation, Data curation. **Jiří Ryšavý:** Writing – review & editing. **Karel Borovec:** Supervision, Resources, Funding acquisition. **Jerzy Górecki:** Writing – review & editing, Supervision, Resources, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

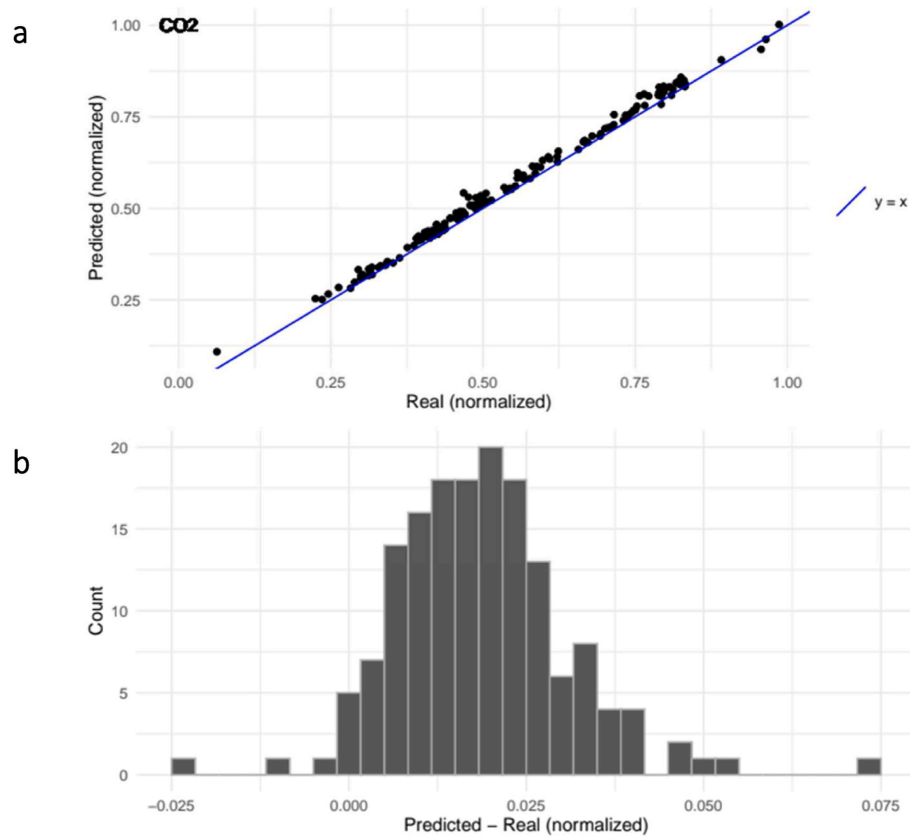


**Fig. 8.** The effectiveness of the predictive features of the ANN model in predicting $CO_2$ content in flue gas is demonstrated in the form of (a) a scatterplot of predicted vs. CO2 measured and (b) a histogram of the difference between predicted and real $CO_2$ concentration values (model topology: inputs/10/4096/128/output).

the work reported in this paper.

## Data availability

Data will be made available on request.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2024.143473.

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